CHE 312 : Chapter 9

1

k α $\hat{C}_p$ q e

1. k, thermal conductivity: the rate of heat that is conducted that is specific to the selected material. Has units of $\frac{W}{m \cdot K}$ $[=]$ $\frac{W}{\text{length} \cdot \text{Temperature}}$

α, thermal diffusivity: $\alpha = \frac{k}{\rho \hat{C}_p}$ $[=]$ $\frac{[\text{length}]^2}{\text{time}}$

$\hat{C}_p$, heat capacity at constant pressure per unit mass: $\hat{C}_p$ $[=]$ $\frac{J}{K \cdot kg}$

q, heat flux: $q_x = -k \frac{dT}{dx}$ or $q = -k \nabla T$ $[=]$ W/m^2

e, combined energy flux: $(\frac{1}{2}\rho v^2 + \rho \hat{U})v + [\pi \cdot v] + q = e$
$[=]$ $\frac{W}{m^2}$ $[=]$ $\frac{kJ}{m^2 \cdot s}$

2. For most solids, k is in the range of $\times 10^2$ to $\times 10^1$. Most liquids are $\times 10^{-1}$. Most gases are $\times 10^{-2}$.

gas liquid solid

increasing k

3. Newton's law

$\tau_{xx} = -\mu \frac{dv_x}{dx}$

- Tensor
- viscosity

$\frac{\Delta \text{ Velocity}}{\text{Displacement}}$

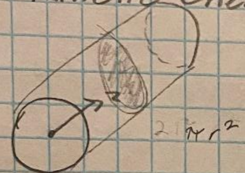
Fourier's Law

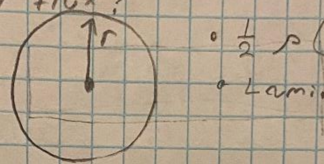
$q_x = -k \frac{dT}{dx}$ $\frac{\Delta \text{ Temp.}}{\text{Displacement}}$

- vector
- Thermal conductivity
- Negative sign next to constants

• $\hat{C}_P - \hat{C}_V = R$ is only true for ideal gases. It should be noted that R is the universal GAS constant. For liquids $Q = m \cdot C_P \cdot \Delta T$ is used.

9. Kinetic energy flux?





• $\frac{1}{2}\rho(v_x^2 + v_y^2 + v_z)v$

• Laminar

$$\frac{1}{2}\rho v^3 \rightarrow \frac{1}{2}\rho\int_0^L\int_0^{2\pi}\int_0^R v^3\, dr\, r\, d\theta\, dz$$